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SWEP 2025 PRACTICE QUESTIONS
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If you notice any error, kindly contact me on
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1. CHEMICAL PROCESS UNIT

1. What is the purpose of PVA in the paint formulation?

- A. Solvent
- B. Binder
- C. Thickener
- D. Additive

Correct Answer: B. Binder. The text explicitly lists PVA as a Binder.

2. Which ingredient is used as a thickener according to the text?

- A. Genitor 1B
- B. FEO
- C. Nitrosol
- D. Industrial soap

Correct Answer: C. Nitrosol. The text identifies Nitrosol as a thickener.

Correct Answer:

3. What is the primary function of the Defoamer?

- A. To add color
- B. To remove foam
- C. To bind ingredients
- D. To make the paint thicker

Correct Answer: B. To remove foam. The text states that Defoamer removes foam.

4. According to the text, which ingredient is classified as a Pigment?

- A. TiO_2
- B. $\text{Al}_2(\text{SO}_3)_3$
- C. Formaldehyde
- D. CaCO_3

B. $\text{Al}_2(\text{SO}_3)_3$. The text explicitly states that $\text{Al}_2(\text{SO}_3)_3$ is a pigment.

5. What does a Binder do in paint?

- A. Provides color
- B. Provides adhesion and strength
- C. Dissolves other ingredients
- D. Removes bad smells

Correct Answer: B. Provides adhesion and strength. The text defines a Binder as providing adhesion, strength, and thickness.

6. In the colorant preparation, what is added to the water first?

Correct Answer:

- A. Defoamer
- B. Industrial soap
- C. Formaldehyde
- D. Premix

Correct Answer: B. Industrial soap. The procedure states to add industrial soap to the water first.

7. How much water is added to the 20 L drum to start with?

- A. 7 L
- B. 50 g
- C. 121 L
- D. 20 L

Correct Answer: A. 7 L. The instructions specify adding 7 L of water to the 20 L drum.

8. According to the text, what are the two main types of paint?

- A. Powder-based and liquid-based
- B. Natural and synthetic
- C. Oil-based and water-based
- D. Soluble and insoluble

C. Oil-based and water-based. The text lists oil-based and water-based as the main types of paint.

9. What ingredient is added to preserve the colorant mix?

Correct Answer:

- A. Genitor 1B
- B. Formaldehyde
- C. Water
- D. Defoamer

Correct Answer: B. Formaldehyde. The procedure notes that you can add formaldehyde to preserve the mix.

10. What is the function of a Surfactant?

- A. It acts as a wetting agent
- B. It gives the paint its color
- C. It provides opacity
- D. It cools the mixture

Correct Answer: A. It acts as a wetting agent. The text lists industrial soap as a surfactant, wetting agent.

11. In the procedures, what ingredient is ground and mixed thoroughly?

- A. TiO_2
- B. $\text{Al}_2(\text{SO}_3)_3$
- C. CaCO_3
- D. Nitrosol

Correct Answer: C. CaCO_3 . The instructions state to grind CaCO_3 and mix it thoroughly.

Correct Answer:

12. What is the first step in the "Flow chart" for making paint?

- A. Adding additives
- B. Getting water in a drum
- C. Mixing TiO_2
- D. Adding industrial soap

B. Getting water in a drum. The flow chart begins with adding water to a 20 L drum.

13. What does a pigment provide to the paint?

- A. A shiny finish
- B. Adhesion and strength
- C. Color and opacity
- D. A way to dissolve ingredients

Correct Answer: C. Color and opacity. The text defines Pigments as providing color and opacity.

14. How long should the solution be stirred nonstop after adding admixtures?

- A. 10 minutes
- B. 20 minutes
- C. 30 minutes
- D. 1 hour

Correct Answer: C. 30 minutes. The procedure specifies stirring for about 30 minutes nonstop.

Correct Answer:

15. Which component is defined as "the actual film-forming component" of paint?

- A.** Pigment
- B.** Solvent
- C.** Binder/Vehicle
- D.** Additive

Correct Answer: C. Binder/Vehicle. The text explicitly defines Binder/Vehicle as the film-forming component.

16. What is added to the mixture after the addition of Nitrosol?

- A.** Water
- B.** Defoamer
- C.** Binder
- D.** Texanol

Correct Answer: C. Binder. After the mixture starts to thicken from the Nitrosol, the next step is to add a binder.

17. Which type of colorant dissolves in oil?

- A. Water soluble
- B. Pigment colorant
- C. Powder form
- D. Oil soluble

Correct Answer: D. Oil soluble. The text explains that oil soluble colorant dissolves only in oil.

18. What is the final step in the colorant preparation?

- A. Adding PVA
- B. Adjusting pH
- C. Stirring to cool
- D. Pouring inside 20 L container

Correct Answer: D. Pouring inside 20 L container. The last step listed in the procedure is to pour the mixture into the container and label it.

19. What is the first step in the procedure for making the solution of TiO_2 and $\text{Al}_2(\text{SO}_3)_3$?

- A. Get a drum of 20 L
- B. Get 2 different containers
- C. Add industrial soap

D. Add water

Correct Answer: B. Get 2 different containers. The procedure begins with the instruction to get two different containers.

20. According to the text, what is the role of the Solvent?

- A. It binds the ingredients.
- B. It removes smells.
- C. It is the water used in the process.
- D. It provides color.

Correct Answer: C. It is the water used in the process. The text lists Water as a Solvent.

2. FOUNDRY UNIT

21. What is a composite material defined as in the text?

- A. A homogeneous mixture of metals.
- B. A single material with superior properties.
- C. A heterogeneous mixture of components.
- D. A mixture made by the hard layer technique.

Correct Answer: C. The text explicitly states that a composite is a heterogeneous mixture of components.

Why the others are wrong: Option A describes an alloy. Option B is incorrect because a composite is a combination of two or more

materials, not a single one. Option D, while a method for creating a composite, is not the fundamental definition provided.

22. In the context of the molding process, what is the purpose of the Vent wire?

- A. To ram the sand.
- B. To sieve the sand.
- C. To hold the core in place.
- D. To provide for gas escape.

Correct Answer: D. The text lists a Vent wire as a tool used in the molding process "for gas escape."

Why the others are wrong: Rammers are used for ramming sand, riddles for sieving, and core prints for holding the core.

23. According to the text, which allowance is used to create a better surface finish on a casting?

- A. Draft allowance
- B. Shrinkage allowance
- C. Machining allowance
- D. Rapping / Shaking allowance

Correct Answer: C. The text defines Machining allowance as being "to get better surface finish."

Why the others are wrong: Draft allowance refers to a taper, shrinkage allowance accounts for size reduction during cooling, and rapping/shaking allowance is for loosening the pattern.

24. What is the function of the Catalyst in the polyester resin mixture?

- A. It speeds up the solidification process.
- B. It acts as a hardener.
- C. It initiates the hardening process.
- D. It is used in a 10:10:0.25 ratio.

Correct Answer: C. The text explicitly states that the Catalyst "initiates the hardening, solidification process."

Why the others are wrong: Option A is the function of the Accelerator. Option B and D are descriptions of the Hardener and the overall ratio, not the function of the Catalyst.

25. What does a Riser do in the gating system?

- A. It directs molten metal into the mold cavity.
- B. It separates the cope and drag.
- C. It serves as a reservoir for feeding molten metal.
- D. It provides a way for gases to escape.

Correct Answer: C. The text defines Risers as "reservoirs in mould for feeding molten metal."

Why the others are wrong: The sprue, runner, and ingate direct the metal. The parting line separates the cope and drag. Vent holes or wires allow gases to escape.

26. Which of these is a type of composite based on its reinforcement?

- A. Hybrid composite
- B. Metal matrix composite
- C. Ceramic matrix composite
- D. Particulate reinforcement

Correct Answer: D. The text lists Particulate reinforcement as a type of composite based on reinforcement.

Why the others are wrong: Options A, B, and C are all classifications of composites based on their matrix materials, not reinforcement.

27. What is the definition of a Foundry according to the text?

- A. A fabrication process.
- B. A specific type of casting.
- C. A mixture of sand and water.
- D. A component of a molding box.

Correct Answer: A. The text states that a Foundry is a "fabrication process."

Why the others are wrong: Options B, C, and D are related concepts but do not define a foundry as a whole.

28. What is the purpose of the Pin in the molding process?

- A. To ram the sand.
- B. To align the cope and drag.
- C. To provide for gas escape.
- D. To strike off excess sand.

Correct Answer: B. The text lists a Pin as a tool used "to align cope + drag." Why the others are wrong: A rammer is used for ramming, a vent wire for gas escape, and a strike rod for striking off excess sand.

29. Which of the following is considered a casting defect?

- A. Machining allowance
- B. Draft allowance
- C. Shrinkage allowance
- D. Misrun

Correct Answer: D. The text provides a list of Casting Defects, and Misrun is included in that list.

Why the others are wrong: Options A, B, and C are all types of allowances, not defects.

30. What is the function of Bentonite in the context of mold materials?

- A. It is a refractory clay.
- B. It is a binder.
- C. Both A and B.

D. It is used to make an engine block.

Correct Answer: C. The text describes Bentonite as a mold material that is a "binder, refractory clay."

Why the others are wrong: Option A and B are both true, but Option C is the most complete and correct answer as it includes both definitions provided in the text. Option D is a casting component, not a property of Bentonite.

31. What is the relationship between Alloy and Composite?

- A.** An alloy is a heterogeneous mixture, while a composite is homogeneous.
- B.** Both are homogeneous mixtures.
- C.** An alloy is a homogeneous mixture of metals, and a composite is a heterogeneous mixture of components.
- D.** Both are made by the hard layer technique.

Correct Answer: C. The text provides the definitions: Alloy = "homogeneous mixture of metals" and Composite = "heterogeneous mixture of components."

Why the others are wrong: Options A and B get the definitions and their properties wrong. Option D describes the process for a composite, not an alloy.

32. What is the purpose of a Core print?

- A.** To provide a parting line.

- B.** To align the cope and drag.
- C.** To hold the core inside the mould.
- D.** To create a replica of the component.

Correct Answer: C. The text defines a Core print as a feature of a pattern used "to hold the core inside mould."

Why the others are wrong: The parting line separates the cope and drag, pins align them, and a pattern is a replica of a component.

33. Which of the following is NOT a type of casting allowance?

- A.** Machining allowance
- B.** Shrinkage allowance
- C.** Hot tears
- D.** Draft allowance

Correct Answer: C. Hot tears are listed under the heading "Casting Defects." All other options are listed under "Allowances."

Why the others are wrong: Machining, Shrinkage, and Draft are all explicitly listed as types of allowances in the text.

34. What is a Riddle used for in the moulding process?

- A.** Mixing sand.
- B.** Sieving sand.
- C.** Ramming sand.
- D.** Making vent holes.

Correct Answer: B. The text lists a Riddle as a tool used "for sieving sand."

Why the others are wrong: A shovel is used for mixing, a rammer for ramming, and a vent wire for vent holes.

35. What is the definition of Stiffness?

- A. The ability to resist deformation.
- B. The energy absorbed before failure.
- C. Resistance to scratching.
- D. The load required to break the material.

Correct Answer: A. The text defines Stiffness as the "ability to resist deformation."

Why the others are wrong: Option B defines toughness, Option C defines hardness, and Option D defines strength.

36. What is the main component of a foundry that is mentioned as a product?

- A. A moulding box
- B. An engine block
- C. A casting
- D. A pattern

Correct Answer: B. The text mentions "Engine block → main component of foundry."

Why the others are wrong: The other options are tools, processes, or parts of the process, but the engine block is specifically mentioned as a product or outcome.

37. Which of the following is a component of moulding sand?

- A. Polyester resin
- B. Silica sand
- C. Egg shell powder
- D. Bentonite

Correct Answer: B. The text lists Silica sand as a component of moulding sand.

Why the others are wrong: Polyester resin and egg shell powder are part of the composite mixture, not moulding sand. While Bentonite is a binder, the prompt asks for a component of moulding sand, and silica sand is the most fundamental component listed.

38. What is the purpose of a Core?

- A. To create the external shape of the casting.
- B. To hold the pattern inside the mould.
- C. To create internal cavities in the casting.
- D. To align the cope and drag.

Correct Answer: C. The text mentions a Core print is used to hold the core, and a core itself is used to create internal cavities, a common foundry term even if not explicitly defined. The context of a "core print" strongly implies the presence of a core for internal features.

Why the others are wrong: A pattern creates the external shape, a core print holds the core, and pins align the cope and drag.

39. What is the first step in the Split pattern moulding process?

- A. Pouring the molten metal.
- B. Turning the drag upside down.
- C. Placing the pattern inside the drag.
- D. Cutting the gating system.

Correct Answer: B. The procedure for the split pattern moulding process begins with "Turn drag upside down."

Why the others are wrong: The other steps, such as placing the pattern, cutting the gating system, and pouring the metal, all happen later in the process.

40. Which of the following is a common application for composite materials mentioned in the text?

- A. Casting
- B. Foundry
- C. Automotive
- D. Moulding

Correct Answer: C. The text lists Automotive as an application for composite materials.

Why the others are wrong: The other options are all related to the casting and foundry processes, not applications for the finished composite materials.

3. PLUMBING, CONCRETE AND BRICKLAYING (PCB) UNIT.

41. What is the definition of "Fresh concrete" according to the text?

- A. Concrete that is already set.
- B. Concrete that can withstand load.
- C. Freshly prepared concrete.
- D. Concrete that has been compacted.

Correct Answer: C. The text states, "Fresh concrete and green concrete = freshly prepared concrete."

Why the others are wrong: Options A and B both describe hardened concrete, not fresh concrete. Option D is a step in the concrete process, but not its definition.

42. Which of the following is considered a lightweight aggregate?

- A. Gravel
- B. Granite
- C. Sawdust
- D. Crushed stone dust

Correct Answer: C. The text lists "sawdust" as a lightweight aggregate used for concrete and road construction.

Why the others are wrong: Gravel and granite are listed as standard coarse aggregates. Crushed stone dust is a fine aggregate.

43. What is the primary use of Sharp sand?

- A. For plastering and rendering.
- B. As a substitute for quarry dust.
- C. To provide high strength to structures.
- D. For mortar and concrete.

Correct Answer: D. The text states, "Sharp sand → preferred for mortar & concrete."

Why the others are wrong: Soft/Plaster sand is used for plastering. Quarry dust is a substitute for fine aggregates, not the primary use of sharp sand. Providing high strength is a property of concrete, not just the sand.

44. In plumbing, what does a Reducer do?

- A. Connects pipes of the same size.
- B. Joins two pipes around a bend.
- C. Regulates and controls flow.
- D. Connects different pipe sizes.

Correct Answer: D. The text states that a Reducer "reduces pipe size" and "connects different sizes."

Why the others are wrong: Adapters connect pipes around a bend. A globe valve regulates and controls flow.

45. What is the primary function of a Ball valve?

- A. To prevent backflow.
- B. To regulate flow.
- C. To prevent leakage.
- D. To connect different pipes.

Correct Answer: C. The text lists "Ball valve, Washers, Teflon tape" as items used "to prevent leakage."

Why the others are wrong: A check valve prevents backflow. A gate valve regulates flow. Adapters and fittings are used to connect pipes.

46. Which of these is a component of a standard mortar mixture?

- A. Coarse aggregate
- B. Cement
- C. Water
- D. All of the above

Correct Answer: B. The text defines mortar as a "mixture of sand + binder (lime, cement, gypsum)." The binder is a key component. While mortar also includes water and fine aggregate (sand), cement is the only option listed that is a core, standard component from the text's definition.

Why the others are wrong: Coarse aggregate is for concrete. Water is not listed as an option but is part of the mix. Cement is the most accurate choice among the options.

47. What is the standard mix ratio for M20 grade concrete?

- A. 1 : 1.5 : 3
- B. 1 : 2 : 4
- C. 1 : 1 : 2
- D. 1 : 3 : 6

Correct Answer: B. The text explicitly lists $M20 = 1 : 2 : 4$.

Why the others are wrong: M15 is 1:2:4, M25 is 1:1.5:3, and M30 is 1:1:2. M10 is 1:3:6.

48. What is the purpose of a Foot valve?

- A. It stops backflow and is fitted inside a well.
- B. It connects pipes to a tank.
- C. It allows for the separation of pipes.
- D. It reduces pipe diameters.

Correct Answer: A. The text states a Foot valve "stops backflow, fitted inside well" and also "at pump inlet."

Why the others are wrong: A backnut allows joining to tanks. A union joint allows for the separation of pipes. Bushing reduces pipe diameters.

49. Which of the following is a type of mortar bond used for brickwork?

- A. Stretcher bond
- B. Header bond
- C. English bond
- D. All of the above

Correct Answer: D. The text lists Stretcher bond, Header bond, English bond, and Flemish bond as types of bonds. Therefore, all the options are correct types of bonds.

Why the others are wrong: Options A, B, and C are all correct, but D is the most complete and accurate answer, as it includes all the listed correct bonds.

50. What is the function of a Trowel in concrete work?

- A. To compact the concrete.
- B. To mix the concrete.
- C. To smooth exposed edges.
- D. To level the concrete.

Correct Answer: C. The text specifies that a Trowel is used "to smooth exposed edges."

Why the others are wrong: A rammer is for compacting, a shovel is for mixing, and a smoother/hammer is for leveling.

51. What is the primary difference between Plain concrete and Reinforced concrete?

- A. Plain concrete is for roads; reinforced concrete is for bridges.
- B. Reinforced concrete has steel bars for tensile strength.
- C. Plain concrete is for slabs; reinforced concrete is for beams.
- D. Plain concrete is for foundations; reinforced concrete is for roads.

Correct Answer: B. The text states that Reinforced Concrete has "steel bars embedded inside" to provide "tensile strength."

Why the others are wrong: The other options describe general uses, but they don't capture the fundamental difference in composition as described in the text.

52. What is the purpose of a Backnut?

- A. To reduce pipe diameters.
- B. To close the ends of pipes.
- C. To allow joining of pipes to tanks.
- D. To prevent backflow.

Correct Answer: C. The text lists a Back nut as something that "allows joining of pipes to tanks."

Why the others are wrong: A bushing reduces diameters, a plug closes pipe ends, and a foot valve prevents backflow.

53. What is the purpose of the Rammer tool?

- A. To mix materials.
- B. To measure quantities.
- C. To compact concrete into position.
- D. To level a surface.

Correct Answer: C. The text defines a Rammer as a tool used to "compact concrete into mould /position."

Why the others are wrong: A shovel is used for mixing, a measuring jar for measuring, and a smoother/hammer for leveling.

54. What is the difference between a Stretcher bond and a Header bond?

- A. Stretcher bond uses bricks laid widthwise, while header bond uses bricks laid lengthwise.
- B. Both use a combination of bricks laid lengthwise and widthwise.
- C. Stretcher bond uses bricks laid lengthwise, while header bond uses bricks laid widthwise.
- D. Stretcher bond uses alternate headers and stretchers in each course.

Correct Answer: C. The text specifies: "Stretcher bond → bricks laid lengthwise" and "Header bond → bricks laid widthwise."

Why the others are wrong: Option A is the reverse of the correct definition. Option B describes both types but not their specific difference. Option D describes a Flemish bond.

55. Which concrete grade is equivalent to a 1 : 2 : 4 mix ratio?

- A. Grade 25
- B. Grade 20
- C. Grade 30
- D. Grade 15

Correct Answer: B. The text lists "1 : 2 : 4 = grade 20."

Why the others are wrong: The text provides the ratios for grades 25, 30, and 15, which are 1:1.5:3, 1:1:2, and 1:2:4, respectively.

56. What is the name of the horizontal layer of bricks in brickwork?

- A. Bond
- B. Bat
- C. Course
- D. Mortar joint

Correct Answer: C. The text defines a Course as a "horizontal layer of bricks."

Why the others are wrong: A bond is the arrangement of bricks, a bat is a cut brick, and a mortar joint is the gap filled with mortar.

57. What is a Tee used for in plumbing?

- A. A 2-way connection.
- B. A 3-way connection.
- C. To regulate flow.
- D. To close the end of a pipe.

Correct Answer: B. The text defines a Tee as a "3-way connection."

Why the others are wrong: A globe valve regulates flow, and a plug closes the end of a pipe.

58. What is the purpose of a Union joint?

- A. To prevent backflow.
- B. To connect pipes of different sizes.
- C. To allow separation of pipes.
- D. To regulate flow.

Correct Answer: C. The text defines a Union joint as something that "allows separation of pipes."

Why the others are wrong: A foot valve prevents backflow, a reducer connects different pipe sizes, and a gate valve regulates flow.

59. What is the purpose of a Vent pipe?

- A. To prevent leakage.
- B. To join pipes to tanks.
- C. To prevent backflow.
- D. It's fitted at the wall top.

Correct Answer: D. The text states that a Vent pipe is "fitted at wall top." While its purpose is to allow air in and out for proper drainage, the text only provides the location.

Why the others are wrong: The other options describe the functions of ball valves, back nuts, and foot valves, not a vent pipe.

60. Which of the following is a type of aggregate used for mortar and concrete?

- A. Gravel
- B. Sand
- C. Crushed stone dust
- D. All of the above

Correct Answer: B. The text defines mortar as a mixture of sand and a binder, and concrete as a mixture of cement, sand, and coarse aggregate. Therefore, sand is a key aggregate for both.

Why the others are wrong: Gravel is only a coarse aggregate used for concrete. While crushed stone dust can be a substitute for sand, sand is the primary component mentioned for both mortar and concrete. Therefore, sand is the most accurate and direct answer.

4. MECHANICAL, POWER AND AUTOMOBILE (MPA) UNIT.

61. Based on the text, what is the purpose of the Catalytic converter in a vehicle's exhaust system?

- A. To remove burnt gases.
- B. To monitor the air-fuel ratio.
- C. To provide power to the wheels.
- D. The text does not specify its purpose.

Correct Answer: D. The text lists the catalytic converter as a component of the exhaust system, but it doesn't provide a specific function for it, unlike the muffler (which removes burnt gases) or the oxygen sensor (which monitors the air-fuel ratio).

Why the others are wrong: Option A is the function of the muffler. Option B is the function of the Oxygen sensor. Option C is the role of the drive and transmission system.

62. In a petrol vehicle's electrical and ignition system, what is the role of the Distributor?

- A. To provide power for ignition.
- B. To ignite the fuel mixture.
- C. To distribute electrical power.
- D. To convert mechanical energy to electrical energy.

Correct Answer: C. The text explicitly states, "Distributor distributes electrical power."

Why the others are wrong: The battery provides power for ignition, the spark plug ignites the fuel, and the alternator converts mechanical energy to electrical energy.

63. What is the function of the Torque converter in an automatic transmission system?

- A. It provides a way to select gears manually.
- B. It replaces the clutch, acting as a hydraulic coupling.

- C. It provides the highest torque at low speed.
- D. It allows the system to decide gear changes automatically.

Correct Answer: B. The text states that a "Torque converter replaces clutch (hydraulic coupling)" in an automatic transmission.

Why the others are wrong: The gear lever is used for manual selection. The gearbox provides torque/speed ratios. The entire automatic transmission system decides gear changes automatically.

64. According to the text, which of these is NOT a function of the lubrication oil?

- A. Reducing friction.
- B. Cooling the engine.
- C. Acting as a detergent.
- D. Transmitting power to the wheels.

Correct Answer: D. The text lists the drive and transmission system as responsible for transmitting power to the wheels. The functions of lubrication oil are listed as cooling, reducing friction, and cleaning (acting as a detergent).

Why the others are wrong: Options A, B, and C are all explicitly listed as functions of lubrication oil in the text.

65. Which type of sensor senses a physical quantity and converts it to electrical energy?

- A. Transducer.

- B. Actuator.
- C. Sensor.
- D. Pneumatics.

Correct Answer: C. The text defines a Sensor as a "device that sense external stimulus (physical quantity) and convert it to electrical energy."

Why the others are wrong: A transducer converts one form of energy to another. An actuator acts as an output device. Pneumatics is a system operating by natural phenomena, not a device that converts physical quantity to electrical energy.

66. In the context of vehicle maintenance, what is the difference between preventive and corrective maintenance?

- A. Preventive maintenance is done when faults exist, while corrective is done at intervals.
- B. Preventive is done at intervals to prevent faults, while corrective is done after faults exist.
- C. Both are done after a fault occurs.
- D. The text does not specify the difference.

Correct Answer: B. The text defines Preventive maintenance as being "carried out at intervals to prevent faults," and Corrective maintenance as being "done when faults already exist."

Why the others are wrong: Option A is the reverse of the correct definitions. Option C is only true for corrective maintenance. Option D is incorrect as the text clearly defines both.

67. What component in a cooling system is responsible for removing heat from the engine to prevent overheating?

- A. The thermostat.
- B. The water pump.
- C. The radiator.
- D. The fan.

Correct Answer: C. The text states that the cooling system "removes heat from engine," and the radiator is the primary component listed for this function.

Why the others are wrong: The thermostat regulates the flow of coolant, the water pump circulates the coolant, and the fan helps the radiator cool down, but the radiator is the main heat exchanger that removes the heat.

68. According to the text, what is the definition of Autotropic?

- A. The combination of mechanical and electronic engineering.
- B. The study of machines that generate their own power.
- C. The application of artificial materials to machines.
- D. The text does not provide a definition.

Correct Answer: D. The text does not provide a direct definition for "Autotropic." It is mentioned alongside Mechatronics, but only the definition for Mechatronics is provided.

Why the others are wrong: Options A, B, and C are incorrect as they are not found in the text or are definitions of other terms.

69. Which of these is a component of a diesel engine listed in the text, but not a petrol engine?

- A. Piston.
- B. Connecting rod.
- C. Glow plug.
- D. Crankshaft.

Correct Answer: C. The text notes a key difference between the two by stating, "spark plug (for petrol), glow plug (for diesel)."

Why the others are wrong: Pistons, connecting rods, and crankshafts are fundamental components of both petrol and diesel engines and are listed under the Diesel Vehicle Maintenance Unit section.

70. What is the function of the Expansion valve in a refrigeration system?

- A. To absorb heat.
- B. To compress the refrigerant.
- C. To release heat.
- D. To reduce pressure and cool the refrigerant.

Correct Answer: D. The text states that the Expansion valve "reduces pressure, cools refrigerant."

Why the others are wrong: The evaporator absorbs heat, the compressor compresses the refrigerant, and the condenser releases heat.

71. Which of the following is a type of sensor that works by operating with natural phenomena like heat and pressure?

- A. Hydraulic.
- B. Pneumatic.
- C. Actuator.
- D. Transducer.

Correct Answer: B. The text states that "Pneumatics → operating by natural phenomena (e.g. heat, pressure)."

Why the others are wrong: Hydraulic systems are more powerful than pneumatic systems but are not defined this way. Actuators are output devices. Transducers convert one form of energy to another.

72. What is the primary function of a vehicle's Fuel System?

- A. To remove burnt gases.
- B. To transmit power to the wheels.
- C. To store and supply fuel.
- D. To provide high speed and low torque.

Correct Answer: C. The text defines the Fuel System as something that "Stores & supplies fuel."

Why the others are wrong: The exhaust system removes burnt gases. The drive and transmission system transmits power to the wheels. High speed and low torque are characteristics of higher gears.

73. What is the purpose of the PTO (Power Take-Off) on a tractor?

- A. To couple implements using a 3-point hitch.
- B. To transmit power to implements.
- C. To pull a plough or ridger.
- D. To provide comfort and absorb shocks.

Correct Answer: B. The text defines PTO as something that "transmits power to implements."

Why the others are wrong: A 3-point hitch is used for coupling implements, a plough is an implement, and suspension absorbs shocks and provides comfort.

74. How much heat does the cooling system remove from the engine, according to the text?

- A. 10%
- B. 50%
- C. ~30%
- D. It does not specify.

Correct Answer: C. The text explicitly states that the cooling system "Removes ~30% of heat."

Why the others are wrong: The text provides a specific percentage, making all other options incorrect.

75. What is the definition of a Transducer given in the text?

- A. A device that senses an external stimulus.
- B. A device that converts one form of energy to another.

- C.** A device that acts as an output.
- D.** A device that provides electrical power.

Correct Answer: B. The text defines a Transducer as a "device that convert one form/domain of energy to another."

Why the others are wrong: Option A defines a sensor, Option C defines an actuator, and Option D is the function of an alternator or battery.

76. What is the primary function of the Thermostat in the cooling system?

- A.** To pump coolant through the engine.
- B.** To store the coolant.
- C.** To regulate the coolant's temperature.
- D.** The text does not specify its function.

Correct Answer: D. While the thermostat's primary function is to regulate temperature, the text only lists it as a component of the cooling system. It does not describe its specific role, unlike the water pump (circulates) or radiator (removes heat).

Why the others are wrong: Options A, B, and C are functions of a water pump, radiator, and the thermostat itself, but the text provided does not explicitly state the thermostat's function.

77. Which type of maintenance is carried out at intervals to prevent faults from occurring?

- A.** Corrective maintenance.

- B.** Preventive maintenance.
- C.** Emergency maintenance.
- D.** Both A and B.

Correct Answer: B. The text defines Preventive maintenance as being "carried out at intervals to prevent faults."

Why the others are wrong: Corrective maintenance is done after a fault has occurred. Emergency maintenance is not mentioned.

78. According to the text, what is the difference between a Dependent and an Independent suspension system?

- A.** Dependent suspension absorbs shocks, while independent provides comfort.
- B.** Independent suspension supports the load, while dependent prevents tire wear.
- C.** In independent suspension, each wheel is separate, while in dependent, the axle supports the load.
- D.** The text does not specify the difference.

Correct Answer: C. The text states, "Independent suspension → each wheel independent. Dependent suspension → axle supports load."

Why the others are wrong: Options A and B describe functions of the suspension system in general, not the difference between the two types. The difference is clearly specified in the text.

79. What is the purpose of the Oil sump in the lubrication system?

- A.** To cool the engine.

- B.** To circulate the oil.
- C.** To check the oil level.
- D.** To store the oil.

Correct Answer: D. The text states that the "Oil sump stores oil."

Why the others are wrong: The oil pump circulates the oil, the lubrication system cools the engine, and the dipstick is used to check the oil level.

80. Which of the following is an example of a Resistive sensor given in the text?

- A.** A Hall effect sensor.
- B.** A thermocouple.
- C.** A fuel gauge.
- D.** A pressure sensor.

Correct Answer: C. The text gives the "fuel gauge" as an example of a Resistive sensor.

Why the others are wrong: A Hall effect sensor is a magnetic sensor. A thermocouple is a temperature sensor. A pressure sensor can be piezoelectric, capacitive, or inductive, but not a resistive sensor according to the text.

5. METAL MACHINING AND FITTING/WELDING FABRICATION (MMF) UNIT.

81. Which type of spanner is listed as being adjustable?

- A. Ring spanner
- B. Socket spanner
- C. Adjustable spanner
- D. The text does not specify

Correct Answer: C. The text lists a few spanner types, including the adjustable spanner.

Why the others are wrong: Ring and socket spanners are listed as types of spanners, but they're not explicitly stated as adjustable. The text directly confirms that the adjustable spanner is, well, adjustable.

82. What is the purpose of a Fuse in an electrical system?

- A. To step up or step down voltage.
- B. To store charge.
- C. To protect against overload.
- D. To oppose current.

Correct Answer: C. The text defines the function of a Fuse as protecting against overload.

Why the others are wrong: A transformer steps up/steps down voltage. A capacitor stores charge. A resistor opposes current.

83. What is the function of a Try square?

- A. To hold a workpiece for marking.
- B. To cut a workpiece.
- C. To check the squareness of a workpiece.
- D. To make internal threads.

Correct Answer: C. The text lists a Try square as a tool used to "check squareness of workpiece."

Why the others are wrong: A magnetic V-block holds a workpiece for marking. A hacksaw is for cutting. A tap and die set is for making threads.

84. What is the primary function of a Transistor?

- A. It stores energy in a magnetic field.
- B. It allows current in one direction.
- C. It is used in amplification.
- D. It stores charge.

Correct Answer: C. The text explicitly states that a Transistor is "used in amplification."

Why the others are wrong: An inductor stores energy in a magnetic field. A diode allows current in one direction. A capacitor stores charge.

85. Which of these machines is used to remove metal using rotary cutters?

- A. Lathe Machine
- B. Drilling Machine

- C. Shaping Machine
- D. Milling Machine

Correct Answer: D. The text defines a Milling Machine as one that "Removes metal using rotary cutters."

Why the others are wrong: A lathe rotates workpieces, a drilling machine makes holes, and a shaping machine is used for shaping but not with rotary cutters.

86. What is the primary function of a Relay?

- A. It is a switching device.
- B. It steps up or steps down voltage.
- C. It protects against overload.
- D. It opposes current.

Correct Answer: A. The text lists a Relay as a "switching device."

Why the others are wrong: A transformer steps up/steps down voltage. A fuse protects against overload. A resistor opposes current.

87. What is a Hacksaw blade used for?

- A. Holding a workpiece.
- B. Making threads.
- C. Bending sheet metal.
- D. Cutting.

Correct Answer: D. The text lists a Hacksaw blade as a tool "for cutting."

Why the others are wrong: A clamp holds a workpiece. A tap and die set makes threads. A bending machine bends sheet metal.

88. According to the text, what is the purpose of a Power supply in a lab?

- A. It generates signals.
- B. It measures voltage and current.
- C. It provides power.
- D. The text does not specify its purpose.

Correct Answer: D. The text lists the Power supply as a common lab equipment but does not define its function, unlike other equipment like the signal generator or multimeter.

Why the others are wrong: A signal generator generates signals. A multimeter measures voltage and current. While a power supply does provide power, the text doesn't explicitly state this function.

89. What is the function of an IC (Integrated Circuit)?

- A. It opposes current.
- B. It is a switching device.
- C. It stores charge.
- D. It is used in control stages

Correct Answer: D. The text defines the function of an IC as being "used in control stages."

Why the others are wrong: A resistor opposes current. A relay is a switching device. A capacitor stores charge.

90. What is the function of a Diode?

- A. It opposes current.
- B. It stores charge.
- C. It allows current in one direction.
- D. It steps up/steps down voltage.

Correct Answer: C. The text states that a Diode "allows current in one direction."

Why the others are wrong: A resistor opposes current. A capacitor stores charge. A transformer steps up/steps down voltage.

91. Which operation is performed on a Lathe Machine?

- a. Face milling
- b. Slotting
- c. Turning
- d. Gear cutting

Correct Answer: C. The text lists turning as an operation performed on a Lathe Machine.

Why the others are wrong: Face milling, slotting, and gear cutting are all operations for a Milling Machine.

92. What does a Distribution Board with a 250 A rating provide?

- A. 6 ways
- B. 9 ways
- C. 18 ways
- D. 36 ways

Correct Answer: B. The text states that a "250 A" DB has "9 ways (standard)."

Why the others are wrong: 200 A is 6 ways, 400 A is 18 ways, and 630 A is 36 ways.

93. Which tool is used to make internal and external threads?

- A. Tap and die set
- B. Pulley extractor
- C. Hand drill machine
- D. Bench grinder

Correct Answer: A. The text states that a Tap and die set is used "to make internal & external threads."

Why the others are wrong: A pulley extractor removes pulleys. A hand drill makes holes. A bench grinder is for grinding.

94. What is the purpose of a Magnetic V-block?

- A. To make internal threads.
- B. To remove pulleys.
- C. To hold a workpiece for marking.
- D. To make a hole.

Correct Answer: C. The text states that a Magnetic V-block is a marking tool used to "hold workpieces."

Why the others are wrong: A tap and die set makes threads. A pulley extractor removes pulleys. A hand drill makes holes.

95. What is the function of an Inductor?

- A. It opposes current.
- B. It stores energy in a magnetic field.
- C. It allows current in one direction.
- D. It is a switching device.

Correct Answer: B. The text states that an Inductor "stores energy in magnetic field."

Why the others are wrong: A resistor opposes current. A diode allows current in one direction. A relay is a switching device.

96. What is the purpose of a Pulley extractor? A. To make threads.

- B. To make holes.**
- C. To remove pulleys.
- D. To hold workpieces.

Correct Answer: C. The text defines a Pulley extractor as a tool "to remove pulleys."

Why the others are wrong: A tap and die set makes threads. A hand drill machine makes holes. A clamp holds workpieces.

97. Which operation is performed on a Milling Machine?

- A. Turning
- B. Boring
- C. Knurling
- D. D. Slotting

Correct Answer: D. The text lists slotting as an operation for a Milling Machine.

Why the others are wrong: Turning, boring, and knurling are all listed as operations for a Lathe Machine.

98. Which of the following is a type of Drilling Machine mentioned in the text?

- A. CNC
- B. Shaping
- C. Radial
- D. Lathe

Correct Answer: C. The text lists the Radial drilling machine as a type of drilling machine.

Why the others are wrong: CNC is a type of milling machine. Shaping and Lathe are separate machines.

99. What is the purpose of a Clamp?

- A. To hold a workpiece for cutting or filing.
- B. To remove pulleys.
- C. To make internal threads.
- D. To make external threads.

Correct Answer: A. The text lists a Clamp as a tool "to hold workpiece for cutting or filing."

Why the others are wrong: A pulley extractor removes pulleys. A tap and die set makes both internal and external threads.

100. According to the text, what is the power factor (P.F) range for a load test?

- A. 0.85 – 0.90
- B. 0.95 – 0.98
- C. 0.70 – 0.80
- D. 0.50 – 0.60

Correct Answer: B. The text specifies that the P.F. for a load test is "0.95 – 0.98."

Why the others are wrong: The other ranges are not mentioned in the text for this specific test.

6. ELECTRICAL POWER AND ELECTRONICS (EPE) UNIT.

101. Based on the text, what is the purpose of the Rectifier diode RM4007 in the 5V Power Supply Circuit?

- A. It stabilizes the voltage.
- B. It converts AC to DC.
- C. It smooths the DC current.
- D. The text does not specify its function.

Correct Answer: D. The text lists the rectifier diode as a component but does not explicitly state its function, unlike other components whose functions are specified.

Why the others are wrong: The voltage regulator stabilizes voltage. The capacitor smooths DC. The text does not explicitly define what the rectifier diode does, so we cannot assume it converts AC to DC, even if that is its real-world function.

102. Which of these wiring systems is listed for domestic use?

- A. Conduit wiring system
- B. Surface wiring system
- C. Trunking wiring system
- D. All of the above

Correct Answer: B. The text specifically lists the Surface wiring system as for "domestic" use.

Why the others are wrong: The other wiring systems are listed as types of wiring systems, but the text does not specify them for domestic use.

103. What is the function of an Auger tool?

- A.** To peel cables.
- B.** To create holes in a wall for cables.
- C.** To hold small wires.
- D.** To cut armored cable.

Correct Answer: B. The text states an Auger is used "to create holes in the wall for cables."

Why the others are wrong: Pliers and knives peel cables. Long nose pliers hold small wires. A hacksaw cuts armored cable.

104. According to the text, what is a key guideline for electrical safety?

- A.** Connect the neutral conductor to the control.
- B.** Use any tool for the job.
- C.** Remove the cap of the cut-out before making repairs.
- D.** Make sure your connection is not accessible.

Correct Answer: C. The text states as a guideline: "Remove the cap of the cut-out before going on the board to make repairs."

Why the others are wrong: The text says to connect the live conductor to the control and to use the appropriate tools. It also says to make sure your connection is accessible.

105. Which color wire is for the Earth wire in a 13A socket outlet?

- A. Brown
- B. Blue
- C. Green/Yellow
- D. Red

Correct Answer: C. The text explicitly states, "Earth wire → Green/Yellow."

Why the others are wrong: Brown is for the live wire and blue is for the neutral wire. Red is not listed.

106. What is the function of the Electrolytic capacitor in the 5V Power Supply Circuit?

- A. To stabilize the voltage.
- B. To display the voltage status.
- C. To smooth DC.
- D. To limit current.

Correct Answer: C. The text states that the Electrolytic capacitor "smooths DC (removes ripple)."

Why the others are wrong: A voltage regulator stabilizes voltage. An LED displays voltage status. A resistor limits current.

107. What is the purpose of a Scraper tool?

- A.** To check for current.
- B.** To make loops.
- C.** To cut and peel cables.
- D.** To remove insulation from cables.

Correct Answer: D. The text states a Scraper is used "to remove insulation from cables."

Why the others are wrong: A tester checks for current. A timble makes loops. Cutting pliers cut and peel cables.

108. According to the text, what is the definition of Safety?

- A.** The correct use of tools for a job.
- B.** Procedures to prevent hazards to personnel, equipment, and environment.
- C.** Using a tester to check a circuit.
- D.** A system for domestic wiring.

Correct Answer: B. The text defines safety as "Procedures put in place to prevent hazards or avert danger to personnel, equipment & environment."

Why the others are wrong: The other options are examples of safe practices or a type of wiring system, but they are not the definition of safety itself.

109. What is a Junction box used for?

- A. To house switches.
- B. To join cables.
- C. To provide a lamp holder.
- D. To fix cables to a surface.

Correct Answer: B. The text states a Junction box is "used for joining cables."

Why the others are wrong: A switch is an appliance. A lamp holder holds a lamp. Cable clips are used to fix cables.

110. What is the difference between a Short circuit fault and an Open circuit fault?

- A. A short circuit is an identified fault, and an open circuit is not.
- B. A short circuit is a complete circuit, and an open circuit is an incomplete circuit.
- C. The text does not provide a definition for either.
- D. They are both the same type of fault.

Correct Answer: C. The text lists both "Short circuit fault" and "Open circuit fault" but does not provide a definition or explanation for the difference between them.

Why the others are wrong: Options A and B are assumed definitions not found in the text. Option D is incorrect because they are listed as two distinct types of faults.

111. Which of the following is a type of capacitor mentioned in the text?

- A. Electrolytic capacitor
- B. Tantalum capacitor
- C. Both A and B
- D. The text only mentions one type of capacitor.

Correct Answer: C. The text lists both Electrolytic capacitor and Tantalum capacitor as types of capacitors.

Why the others are wrong: Options A and B are correct on their own, but C is the most complete answer. D is incorrect as the text lists two types.

112. Which of the following is an example of an electrical circuit type?

- A. Surface wiring system
- B. Looping circuit
- C. Trunking wiring system
- D. Conduit wiring system

Correct Answer: B. The text lists Looping Circuit as a "Type of Electrical Circuit."

Why the others are wrong: Surface, Trunking, and Conduit are listed as "Types of Wiring Systems," not types of electrical circuits.

113. In the 13A socket outlet, what is the color of the Live wire?

- A. Brown
- B. Blue C. Green/Yellow
- D. Black

Correct Answer: A. The text states, "Live wire → Brown."

Why the others are wrong: Blue is for the neutral wire, and Green/Yellow is for the earth wire. Black is not listed.

114. What is the function of the Voltage regulator in the 5V Power Supply Circuit?

- A. It smooths the DC.
- B. It stabilizes the voltage.
- C. It limits current.
- D. It displays the voltage status.

Correct Answer: B. The text defines a Voltage regulator as a component that "stabilizes voltage."

Why the others are wrong: The capacitor smooths DC. The resistor limits current. The LED displays voltage status.

115. What is a Gauged mortar made of?

- A. Cement and fine aggregate.
- B. Lime and fine aggregate.
- C. Cement and lime.
- D. Sand and binder.

Correct Answer: C. The text mentions Gauged mortar as a type of mortar made from "cement + lime."

Why the others are wrong: Sand and binder are the general components of mortar. Cement is for cement mortar, and lime is for lime mortar.

116. What is a thimble used for?

- A. To remove insulation from cables.
- B. To make loops.
- C. To cut and peel cables.
- D. To check for current.

Correct Answer: B. The text lists a Thimble as a tool used "for making loops." Why the others are wrong: A scraper removes insulation. Cutting pliers cut and peel cables. A tester checks for current.

117. How is the value of a Colour-coded resistor determined?

- A. By its size.
- B. By its wattage.
- C. By its colour bands.
- D. By testing it with a multimeter.

Correct Answer: C. The text states that for a colour-coded resistor, "colour bands determine resistor value."

Why the others are wrong: The size determines the wattage. A multimeter is used to test the resistance value, but the colour bands determine its marked value.

118. Which of the following is a type of Pliers listed in the text?

- A. Long nose pliers
- B. Flat pliers
- C. Short nose pliers
- D. Round nose pliers

Correct Answer: A. The text lists Long nose pliers and Cutting pliers as types of pliers.

Why the others are wrong: Flat, short, and round nose pliers are not listed in the provided text.

119. What is the definition of a Ring Circuit?

- A. A circuit where one line supplies outlets directly.
- B. A circuit where the supply point loops from one outlet to another.
- C. A circuit connected end-to-end cable.
- D. A circuit for power circuits.

Correct Answer: C. The text defines a Ring Circuit as a "circuit connected end-to-end cable."

Why the others are wrong: Option A describes a Radial Circuit. Option B describes a Looping Circuit.

120. What is the purpose of a Teflon tape in plumbing?

- A. To prevent leakage.
- B. To regulate flow.

- C. To prevent backflow.
- D. To connect pipes.

Correct Answer: A. The text lists "Teflon tape" along with ball valves and washers as tools used "to prevent leakage."

Why the others are wrong: A gate valve regulates flow. A check valve prevents backflow. Adapters connect pipes.

7. FOOD PROCESSING AND AEROPONICS (FPA) UNIT.

121. Based on the text, what is the purpose of Pasteurization in yoghurt production?

- A. To add bacteria.
- B. To stop fermentation.
- C. To kill pathogenic microorganisms.
- D. To convert lactose to lactic acid.

Correct Answer: C. The text explicitly states that pasteurization "kills pathogenic microorganisms."

Why the others are wrong: Bacteria are added during inoculation. Pressure and cooling are used to stop fermentation. The bacteria convert lactose to lactic acid during incubation.

122. What is the definition of Milling?

- A. The breakdown of large materials into smaller pieces.

- B.** The use of centrifugal and impact forces.
- C.** The breakdown of agricultural products into powder, paste, or granules.
- D.** A method that uses two rollers rotating in opposite directions.

Correct Answer: C. The text defines Milling as "The breakdown of agricultural products into smaller particles (powder, paste, granules)."

Why the others are wrong: Option A is the definition of "Size Reduction." Option B describes the operation of a ball mill. Option D describes a roller mill.

123. According to the text, which of these is a type of Milling equipment?

- A.** Slicing
- B.** Dapping
- C.** Hammer Mill
- D.** Trimming

Correct Answer: C. The text lists Hammer Mill as an example of a mill.

Why the others are wrong: Slicing, dapping, and trimming are listed as general methods of size reduction, not specific types of mills.

124. What is a key characteristic of Vertical Farming?

- A.** Maximizing land use in large spaces.
- B.** Planting crops in a soil medium.
- C.** Planting crops on vertical shelves.

D. Producing crops with traditional farming methods.

Correct Answer: C. The text describes vertical farming as involving "planting crops on vertical shelves.

Why the others are wrong: Vertical farming is used to maximize land use in small spaces and involves a soil-less medium. It is a modernized system, not a traditional one.

125. In the context of the cookies recipe, what ingredient is added last before rolling and baking?

- A.** Flour and baking powder
- B.** Eggs and milk
- C.** Margarine and sugar
- D.** Flavour

Correct Answer: D. The procedure lists adding flavour after folding in the flour and baking powder.

Why the others are wrong: Margarine and sugar are mixed first, followed by eggs and milk, and then the flour and baking powder are folded in.

126. What is the definition of Food?

- A.** A substance that provides energy.
- B.** A substance that can be metabolized by the body.
- C.** A substance used to repair tissues.
- D.** All of the above.

Correct Answer: D. The text defines food as a substance that can be metabolized to "generate energy, build, and repair tissues." All listed options are part of this definition.

Why the others are wrong: Options A, B, and C are all parts of the definition, but D is the most complete and accurate answer.

127. What two bacteria are used as starter cultures in industrial yoghurt production?

- A. Lactobacillus and Streptococcus.
- B. Lactobacillus bulgaricus and Streptococcus thermophilus.
- C. Lactobacillus and Streptococcus thermophilus.
- D. Lactobacillus bulgaricus and Streptococcus.

Correct Answer: B. The text specifically lists Lactobacillus bulgaricus and Streptococcus thermophilus.

Why the others are wrong: Options A, C, and D are not specific enough to match the names given in the text.

128. According to the text, what is Food Processing?

- A. The design of food production machinery.
- B. Any activity to preserve, convert, or improve agricultural raw materials.
- C. A method of processing like baking or roasting.
- D. The preparation of a recipe.

Correct Answer: B. The text defines Food Processing as "Any activity performed on agricultural raw materials to preserve, convert, or improve their value."

Why the others are wrong: Option A is the definition of Food Engineering. Options C and D are examples of specific food processing activities or preparation, but not the overall definition.

129. What is the role of an Accelerator in the Yoghurt Production process?

- A. To initiate hardening.
- B. To remove bacteria.
- C. To provide a better taste.
- D. The text does not mention an accelerator.

Correct Answer: D. The text does not mention an accelerator in the yoghurt production process. This term was present in the previous text about polyester resin.

Why the others are wrong: Options A, B, and C are incorrect based on the provided text, as the term and its function are not present in this document.

130. What is the purpose of inoculation in yoghurt production?

- A. To cool the milk.
- B. To add bacteria to the milk.
- A.

- B.
- C.
- D.

Correct

- C.** To convert lactose to lactic acid.
- D.** To stop fermentation.

Correct Answer: B. The text defines the Inoculation Process as keeping milk in an enclosed space for a period of time at a specific temperature (45 °C), which is the process of adding the bacteria.

Why the others are wrong: Cooling is a separate step before and after inoculation. The bacteria convert the lactose, but that is not the definition of the inoculation process itself. Pressure and cooling stop fermentation.

131. Which of the following is a type of food based on its source?

- A. Carbohydrates
- B. Yoghurt
- C. Plant-based
- D. Roughages

Answer: C. The text classifies food "Based on Sources" into Plant-based, Animal-based, and Synthesized.

Why the others are wrong: Carbohydrates and roughages are classified based on nutrients/functions. Yoghurt is a specific food product, not a source classification.

132. What force is used in the operation of a Ball Mill?

- A.

- B.
- C.
- D.

Correct

- A. Shearing forces.
- B. Compression forces.
- C. Impact forces.
- D. Both A and C.

Correct Answer: C. The text states that a Ball Mill "Uses centrifugal & impact forces." It also reduces particle size by "impact & attrition."

Why the others are wrong: Shearing and compression forces are listed as general principles of milling, but not specifically for a ball mill's operation.

133. What is the definition of Food Engineering?

- A. The processing of agricultural raw materials.
- B. The design of food production machinery and systems.
- C. The study of different types of food.
- D. The process of converting protein to amino acids.

Correct Answer: B. The text defines Food Engineering as dealing with "the design of food production machinery & systems for food processing."

Why the others are wrong: Option A is the definition of Food Processing.

- A.

- B.
- C.
- D.

Correct

Options C and D are related topics but not the definition of food engineering.

134. What is the purpose of Coagulation in yoghurt production?

- A. To break down milk proteins.
- B. To convert lactose into acid.
- C. To form the yoghurt.
- D. The text does not specify the purpose.

Answer: C. The text mentions that acid acts on milk proteins, causing coagulation, which forms the yoghurt.

Why the others are wrong: The acid acts on milk proteins, it does not break them down. Bacteria convert lactose to lactic acid. The text shows the process of coagulation to form the final product.

135. Which of the following is an example of a processed food according to the text?

- A. Grilled/Roasted foods
- B. Condiments/Soups
- C. Cookies
- D. All of the above

Correct Answer: D. The text lists all of these as examples of processed foods and their corresponding production processes.

- A.

- B.
- C.
- D.

Correct

Why the others are wrong: All options are correct examples of processed foods from the list provided, making D the most accurate answer.

136. What is the temperature range for Inoculation?

- A. 65–70 °C
- B. 42–45 °C
- C. 5 °C
- D. 200 °C

Correct Answer: B. The text states the temperature for Inoculation is "42–45 °C."

Why the others are wrong: 65–70 °C is for pasteurization. 5 °C is for cooling after incubation. 200 °C is for baking cookies.

137. What is the first step in the procedure for making cookies?

- A. Add eggs & milk.

A.

- B.
- C.
- D.

Correct

- B. Add flavour.
- C. Mix margarine & sugar until fluffy.
- D. Fold in flour & baking powder.

Answer: C. The procedure lists "Mix margarine & sugar until fluffy" as the first step.

Why the others are wrong: The other options are subsequent steps in the recipe.

138. Which of the following is a type of food based on its Nutrients/Functions?

- A. Plant-based
- B. Animal-based
- C. Synthesized
- D. Fats & Oils

Correct Answer: D. The text lists Fats & Oils under the classification "Based on Nutrients/Functions."

Why the others are wrong: Plant-based, animal-based, and synthesized are classifications based on food sources.

139. What is the purpose of incubation?

- A. To keep the milk in an enclosed space for a period of time.
- B. To convert lactose to lactic acid.

- C. To cool the milk after pasteurization.
- D. To kill pathogenic microorganisms.

Correct Answer: A. The text states that the "Inoculation Process" involves "Keeping milk in an enclosed space for a period of time at 45 °C," which refers to the incubation stage where the inoculated milk is held to ferment.

Why the others are wrong: Conversion to lactic acid is a result of the bacteria's activity during incubation, but not the purpose of the step itself. Cooling is a separate step before and after incubation. Killing microorganisms is the purpose of pasteurization.

140. What is the relationship between Size Reduction and Milling?

- A. They are the same thing.
- B. Milling is a type of size reduction.
- C. Size reduction is a type of milling.
- D. They are completely unrelated processes.

Correct Answer: B. The text lists Milling as a specific example of Size Reduction. The definition of size reduction is broad, while milling is a specific method.

Why the others are wrong: They are not the same thing because size reduction includes methods like slicing and grating. Size reduction is a broader category that includes milling. They are related processes.

8. EQUIPMENT MAINTENANCE AND CENTRE (EMC) UNIT.

141. According to the text, what is a key difference between a Static Fault and a Dynamic Fault?

- A. Static faults are sudden, while dynamic faults are caused by open or short circuits.
- B. Static faults are caused by open circuits, while dynamic faults are caused by short circuits.
- C. Static faults are caused by short circuits, while dynamic faults are caused by open circuits.
- D. The text does not specify the difference.

Correct Answer: A. The text states that static faults are "sudden, includes breakdown," while dynamic faults are "caused by open/short circuits."

Why the others are wrong: Options B and C reverse the correct definitions. Option D is incorrect as the text provides clear definitions for both.

142. What is the function of a Transformer?

- A. It stores charge in a magnetic field.
- B. It allows current to flow in one direction only.
- C. It steps up or steps down AC voltage.
- D. It opposes the flow of current.

Correct Answer: C. The text explicitly defines a Transformer as something that "steps up/steps down AC voltage."

Why the others are wrong: An inductor stores charge in a magnetic field. A diode allows current in one direction. A resistor opposes the flow of current.

143. What is a Short-circuit fault?

- A. A fault where there is no electrical conduction.
- B. An unintended connection to earth ground.
- C. A fault where there is unintended conduction.
- D. A fault caused by a broken component.

Correct Answer: C. The text defines a Short-circuit fault as having "unintended conduction."

Why the others are wrong: An open-circuit fault has no electrical conduction. A ground fault is a connection to earth ground. The text does not define a fault as being caused by a broken component.

144. According to the text, what is the relationship between the size of a pulley and its speed?

- A. The size is directly proportional to the speed.
- B. The size is inversely proportional to the speed.
- C. The size is equal to the speed.
- D. The text does not specify a relationship.

Correct Answer: B. The text states that "Size of pulley is inversely proportional to the speed."

Why the others are wrong: Options A and C are incorrect as they contradict the statement in the text. Option D is incorrect because the relationship is explicitly stated.

145. Which of these is a component of an electric fan?

- A. Thermostat
- B. Carbon brushes
- C. Stator
- D. Commutator

Correct Answer: C. The text lists stator as a component of an electric fan.

Why the others are wrong: A thermostat is in an electric iron and kettle. Carbon brushes and a commutator are components of a blender.

146. What is the purpose of a UPS in a computer system?

- A. To provide input for the computer.
- B. To provide a permanent storage device.
- C. To print documents.
- D. To provide backup power.

Correct Answer: D. The text defines a UPS as providing "backup power."

Why the others are wrong: A keyboard and mouse are input devices. A hard drive is a permanent storage device. A printer prints documents.

147. What does an Integrated Circuit (IC) contain in a single package?

- A. A single transistor.
- B. A single diode and a resistor.
- C. Diodes, resistors, and capacitors.
- D. A transformer.

Correct Answer: C. The text states that ICs "Contain diodes, resistors, capacitors in single package."

Why the others are wrong: The other options are incomplete or incorrect based on the text's description.

148. According to the text, what is the purpose of the Dipstick in a car's engine?

- A. To store oil.
- B. To circulate oil.
- C. To check the oil level.
- D. The text does not mention a dipstick.

Correct Answer: D. The text mentions a dipstick in the previous text about lubrication systems, but it is not present in the current text provided.

Why the others are wrong: Options A, B, and C are functions mentioned in the previous text but are not present in this new text.

149. Which of these is a criterion for choosing the right measuring instrument?

- A. Cost
- B. Appearance
- C. Size
- D. Brand name

Correct Answer: A. The text lists Cost as one of the criteria for choosing an instrument.

Why the others are wrong: Appearance, size, and brand name are not listed as criteria.

150. What is the function of a Transistor?

- A. It opposes the flow of current.
- B. It stores charge.
- C. It is used for switching and amplification.
- D. It allows current to flow in only one direction.

Correct Answer: C. The text defines a Transistor as being used for "switching & amplification."

Why the others are wrong: A resistor opposes current. A capacitor stores charge. A diode allows current in one direction.

151. What are the colors of the Live, Neutral, and Earth wires in an electric iron's plug, respectively?

- A. Brown, Blue, Green
- B. Brown, Green, Blue

- C. Green, Blue, Brown
- D. Brown, Blue, Green/Yellow

Correct Answer: D. The text states, "Brown wire = live, Blue = neutral, Green = earth." It also specifies that the earth wire is green and yellow in a previous text.

This text specifically says Green, but in context of the other text, Green/Yellow is the most accurate answer.

Why the others are wrong: The other options do not match the correct color code provided in the text.

152. Which of the following is a type of fault caused by an unintended connection to earth ground?

- A. Open-circuit fault
- B. Short-circuit fault
- C. Ground fault
- D. Static fault

Correct Answer: C. The text defines a Ground fault as a "connection to earth ground."

Why the others are wrong: An open-circuit fault has no conduction. A short-circuit fault has unintended conduction. A static fault is a sudden breakdown.

153. According to the text, what is a faulty instrument system?

- A. A system with an open circuit.

- B. A system with a short circuit.
- C. A system not able to produce the expected output.
- D. A system with a ground fault.

Correct Answer: C. The text defines a Faulty Instrument System as "A system not able to produce expected output."

Why the others are wrong: Options A, B, and D are specific types of faults, not the general definition of a faulty system.

154. What is the function of a Diode?

- A. It allows current to flow one way only.
- B. It stores charge.
- C. It opposes the flow of current.
- D. It is used for amplification.

Correct Answer: A. The text defines a Diode as something that "allows current one way only."

Why the others are wrong: A capacitor stores charge. A resistor opposes current. A transistor is used for amplification.

155. Which of these is a component of a Blender?

- A. Capacitor
- B. Stator
- C. Carbon brushes
- D. Thermostat

Correct Answer: C. The text lists carbon brushes as a component of a blender.

Why the others are wrong: A capacitor and stator are components of a fan. A thermostat is a component of an iron and kettle.

156. What is the purpose of the Thermostat in an electric iron?

- A. To provide power.
- B. To store charge.
- C. To regulate temperature.
- D. The text does not specify its purpose.

Correct Answer: D. The text lists the thermostat as a component of the iron but does not describe its specific function.

Why the others are wrong: Options A, B, and C are functions of a plug, a capacitor, and the thermostat itself, but the text does not explicitly state the thermostat's function.

157. What is the main function of an Inductor?

- A. It opposes the flow of current.
- B. It stores charge in a magnetic field.
- C. It steps up or steps down AC voltage.
- D. It allows current to flow in one direction.

Correct Answer: B. The text defines an Inductor as something that "stores charge in magnetic field."

Why the others are wrong: A resistor opposes current. A transformer steps up/steps down voltage. A diode allows current in one direction.

158. What is the primary function of a Multimeter in troubleshooting?

- A. To cut wires.
- B. To tighten bolts.
- C. To measure electrical and electronic quantities.
- D. To make threads.

Correct Answer: C. The text lists a Multimeter as a tool for troubleshooting and specifies its function as "electrical/electronic" measurement.

Why the others are wrong: Pliers cut wires. Spanners tighten bolts. A tap and die set makes threads.

159. What is the difference between a Polarized and a Non polarized capacitor?

- A. Polarized capacitors are ceramic, while non-polarized are electrolytic.
- B. Polarized capacitors are electrolytic, while non-polarized are ceramic.
- C. Polarized capacitors store energy, while non-polarized capacitors filter circuits.
- D. The text does not specify the difference.

Correct Answer: B. The text states, "Polarized capacitors → electrolytic" and "Non-polarized capacitors → ceramic."

Why the others are wrong: Option A reverses the correct definitions. Option C describes general uses of a capacitor. Option D is incorrect as the text clearly defines the difference.

160. Which of these is an example of an Output device?

- A. Keyboard
- B. Scanner
- C. Monitor
- D. Flash drive

Correct Answer: C. The text lists a Monitor as an output device.

Why the others are wrong: A keyboard and scanner are input devices. A flash drive is a storage device.

9. BIOMEDICAL TECHNOLOGY (BT)/WOOD AND UPHOLSTERY (WU) UNIT.

161. According to the text, which of the following is a type of Diagnostic Equipment?

- A. Pacemaker
- B. Ventilator
- C. Stethoscope
- D. X-ray machine

Correct Answer: D. The text lists the X-ray machine as an example of diagnostic equipment.

Why the others are wrong: A pacemaker and ventilator are examples of treatment/therapeutic equipment, and a stethoscope is a monitoring device.

162. What is the function of an Electrical safety analyzer?

- A. It checks the power output of surgical devices.
- B. It checks the output of a defibrillator.
- C. It checks for leakage current.
- D. It measures vital signs.

Correct Answer: C. The text states that an Electrical safety analyzer "checks leakage current."

Why the others are wrong: An electrosurgical analyzer checks the power output of surgical devices. A defibrillator tester checks defibrillator output. Vital signs are measured by devices like a stethoscope or thermometer.

163. What is the normal Diastolic pressure for an adult?

- A. 120 mmHg
- B. 80 mmHg
- C. 60 bpm
- D. 16 breaths/min

Correct Answer: B. The text lists the normal BP for adults as 120/80 mmHg, where 80 mmHg is the diastolic pressure.

Why the others are wrong: 120 mmHg is the systolic pressure. 60 bpm is the normal pulse rate. 16 breaths/min is the normal respiration rate.

164. According to the text, which of the following medical devices is classified as Class III?

- A. Wheelchairs
- B. Infusion pumps
- C. Heart valves
- D. Pulse oximeters

Correct Answer: C. The text lists Heart valves as an example of a Class III medical device.

Why the others are wrong: Wheelchairs are Class I. Infusion pumps and pulse oximeters are Class II.

165. What is the definition of Systole in the context of the vital signs simulator?

- A. Relaxation
- B. Contraction
- C. The first sound heard when measuring blood pressure
- D. The last sound heard when measuring blood pressure

Correct Answer: B. The text states, "Systole → contraction."

Why the others are wrong: Diastole is relaxation. The first sound is the systolic pressure, but contraction is the definition of systole itself. The last sound disappearing is the diastolic pressure.

166. Which of these is a property of an Operational Amplifier (Op-Amp)?

- A. Low gain
- B. Very low input impedance
- C. Very high output impedance
- D. Very high input impedance

Correct Answer: D. The text lists "Very high input impedance" as a property of an Op-Amp.

Why the others are wrong: The other options are the opposite of the properties listed: Op-Amps have high gain and very low output impedance.

167. What is the formula for the gain of an Inverting Amplifier?

- A. $A_v = 1 + \frac{R_f}{R_{in}}$
- B. $V_{out} = \frac{R_f}{R_1} (V_2 - V_1)$
- C. $A_v = - \frac{R_f}{R_{in}}$
- D. The text does not provide a formula.

Correct Answer: C. The text provides the formula for an inverting amplifier as $A_v = - \frac{R_f}{R_{in}}$.

Why the others are wrong: Option A is the formula for a non-inverting amplifier, and Option B is the formula for a differential amplifier.

168. What is the range of a normal adult's Respiration rate?

- A. 60 – 100 bpm
- B. 16 – 20 breaths/min
- C. 120/80 mmHg
- D. 36.5 – 37.4 °C

Correct Answer: B. The text lists the normal respiration rate as "16 – 20 breaths/min."

Why the others are wrong: 60-100 bpm is the normal pulse. 120/80 mmHg is the normal blood pressure. 36.5-37.4 °C is the normal body temperature.

169. What is the purpose of Negative Feedback in an Op-Amp circuit?

- A. It increases gain.
- B. It may cause instability.
- C. It stabilizes output.
- D. It amplifies noise.

Correct Answer: C. The text states that Negative Feedback "stabilizes output, improves linearity."

Why the others are wrong: Positive feedback increases gain and may cause instability. Negative feedback reduces common-mode noise.

170. In the procedure for taking blood pressure, what does the last sound disappearing indicate?

- A. Systolic pressure
- B. Diastolic pressure
- C. Normal pressure
- D. A faulty reading

Correct Answer: B. The text states, "Last sound disappears = Diastolic pressure."

Why the others are wrong: The first sound indicates systolic pressure. The other options are incorrect based on the text.

171. Which of the following is a type of Op-Amp connection?

- A. Differential Amplifier
- B. Comparator
- C. Integrator
- D. All of the above

Correct Answer: D. The text lists Differential Amplifier, Comparator, and Integrator as types of Op-Amp connections.

Why the others are wrong: All options are correct types of connections, so D is the most complete answer.

172. Which of the following is an example of a Medical Test Equipment?

- A. Stethoscope
- B. Pacemaker
- C. Defibrillator tester

D. Patient monitor

Correct Answer: C. The text lists the Defibrillator tester as an example of medical test equipment.

Why the others are wrong: A stethoscope is a monitoring device. A pacemaker is a treatment device. A patient monitor is a monitoring device.

173. According to the text, what is the normal range for an adult's Body temperature?

- A. 36.5 – 37.4 °C
- B. 35.0 – 36.0 °C
- C. 37.5 – 38.0 °C
- D. 36.5 – 37.4 °F

Correct Answer: A. The text lists normal body temperature as "36.5 – 37.4 °C."

Why the others are wrong: The other temperature ranges are incorrect, and the unit is degrees Celsius, not Fahrenheit.

174. What is the function of a Defibrillator tester?

- A. To check for leakage current.
- B. To check the defibrillator's output.
- C. To check the power output of surgical devices.
- D. The text does not specify the function.

Correct Answer: B. The text states that a Defibrillator tester "checks defibrillator output."

Why the others are wrong: An electrical safety analyzer checks for leakage current, and an electrosurgical analyzer checks the power output of surgical devices.

175. Which of the following is an example of a Treatment/Therapeutic Equipment?

- A. X-ray machine
- B. Thermometer
- C. Dialysis machine
- D. Glucometer

Correct Answer: C. The text lists the Dialysis machine as a type of treatment/therapeutic equipment.

Why the others are wrong: An X-ray machine is diagnostic. A thermometer and a glucometer are monitoring devices.

176. What is the purpose of a Vital Signs Simulator?

- A. To measure a patient's vital signs.
- B. To treat a patient's condition.
- C. To simulate patient conditions like ECG and BP.
- D. To analyze an ECG wave.

Correct Answer: C. The text states that a Vital Signs Simulator "Simulates patient conditions: ECG, BP, pulse, respiration."

Why the others are wrong: Patient monitors measure vital signs. Therapeutic equipment treats conditions. The simulator can simulate an ECG, but its purpose is broader than just analysis.

177. Which of these is a component of a Blood Pressure Machine used for the practical?

- A. Blood pressure cuff
- B. Stethoscope
- C. Sphygmomanometer
- D. All of the above

Correct Answer: D. The text lists the sphygmomanometer (with its cuff) and the stethoscope as tools for using a blood pressure machine.

Why the others are wrong: All three items are listed as tools used in the procedure, making D the most complete answer.

178. What is the function of the Output of an Op-Amp?

- A. It has differential inputs.
- B. It is the amplified signal.
- C. It has very high input impedance.
- D. It has a high gain.

Correct Answer: B. The text states that the Output is the "amplified signal."

Why the others are wrong: Options A, C, and D are properties of the Op-Amp itself, not a description of its output.

179. What is the purpose of Positive Feedback in an Op-Amp?

- A. To stabilize the output.
- B. To improve linearity.
- C. To increase gain.
- D. To reject common-mode noise.

Correct Answer: C. The text states that Positive Feedback "increases gain, may cause instability."

Why the others are wrong: Negative feedback stabilizes the output and improves linearity. Common-mode rejection is a property of the Op-Amp itself.

180. What is the normal pulse rate for an adult?

- A. 120/80 mmHg
- B. 60 – 100 bpm
- C. 16 – 20 breaths/min
- D. 36.5 – 37.4 °C

Correct Answer: B. The text lists the normal pulse as "60 – 100 bpm."

Why the others are wrong: The other options are normal ranges for blood pressure, respiration, and body temperature, respectively.

If you notice any error, kindly contact me on
08164761492 thanks.

MR MALVEN